Contagion *Graph* / supply-chain contagion, traced from sec filings

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Every headline has *a blast radius.*

> *"Plain RAG retrieves documents; Contagion Graph retrieves consequences."*

Corporate risk does not stop at direct tier-1 counterparties. When a major earthquake halts semiconductor foundries in Taiwan, or trade sanctions restrict lithography tooling, the decisive damage cascades through **2nd- and 3rd-hop dependencies**: substrate suppliers, packaging subcontractors, automotive MCU makers, and consumer hardware integrators across multiple jurisdictions.

- **LLMs Hallucinate Counterparties**
  General generative models invent plausible-sounding suppliers from training co-occurrence. They cannot verify operational contracts or calculate auditable financial impact weights.

- **SEC 10-K Filings Are Opaque**
  Firms bury sole-supplier risks and >10% customer revenue thresholds across thousands of legal pages. Standard keyword RAG retrieves isolated sentences but fails on transitive multi-hop chains.

01 Offline Ingestion & Verified Knowledge Graph

116 SEC Filings
Harvested from SEC EDGAR across 146 tracked corporations.

557 Resolved Entities
411 external supplier nodes linked via fuzzy ticker matching.

Verbatim Quote Rule
Extracted edges must match word-for-word against filing text.

1,224 Active Edges
1,150 filing quotes + 140 curated links; rebuilds cost ~$0.15.

02 Interface: Concentric Radial Shock Diffusion

Figure 1: Main Platform Interface — Natural-language headline search bar, real-time news feed (GDELT / Google News), and Cytoscape.js radial graph demonstrating multi-hop shock propagation outward from TSMC (TSM).

6-Stage Pipeline & *Deterministic Scoring.*

When an event headline is submitted, the online pipeline executes across six structured stages in under 900 milliseconds, ensuring completely reproducible mathematical scoring without generative hallucinations:

1. **Parse Event** — 10 Archetypes (Haiku)
2. **Locate Origin** — Graph Root Map
3. **Trace Paths** — 3 Hops (Top 40/hop)
4. **Score Impact** — Deterministic Math
5. **Explainer** — GraphRAG Citations
6. **Critic** — Verbatim Audit

Directed Traversal Templates

The Navigator follows strict event rules (10 archetypes: supply disruptions, regional disasters, sanctions, commodity shocks). Supply halts walk downstream toward customers; demand cuts walk upstream; competitor edges invert sign (-1).

Adversarial Critic Judge

An autonomous critic compares synthesized narrative claims against the cited statutory sentence. Unsupported extrapolations or links >18 months old are visibly flagged with amber warnings to eliminate hallucinations.

03 Mathematical Scoring Formulation

$$\text{Impact}(c) = S_e \times \text{polarity} \times \sum \left[\prod (w_{uv} \cdot s_{uv}) \right] \times \lambda^{(\text{hops} - 1)} \times f_{\text{fresh}}(p)$$

S_e: Severity (0.2–1.0) • **polarity:** ±1 • **w_uv:** 10-K dependency weight • **s_uv:** +1 customer / -1 competitor • **λ:** Hop decay factor (0.50) • **f_fresh:** Evidence freshness decay (>12mo)

04 Execution Trace: US Export Controls to China

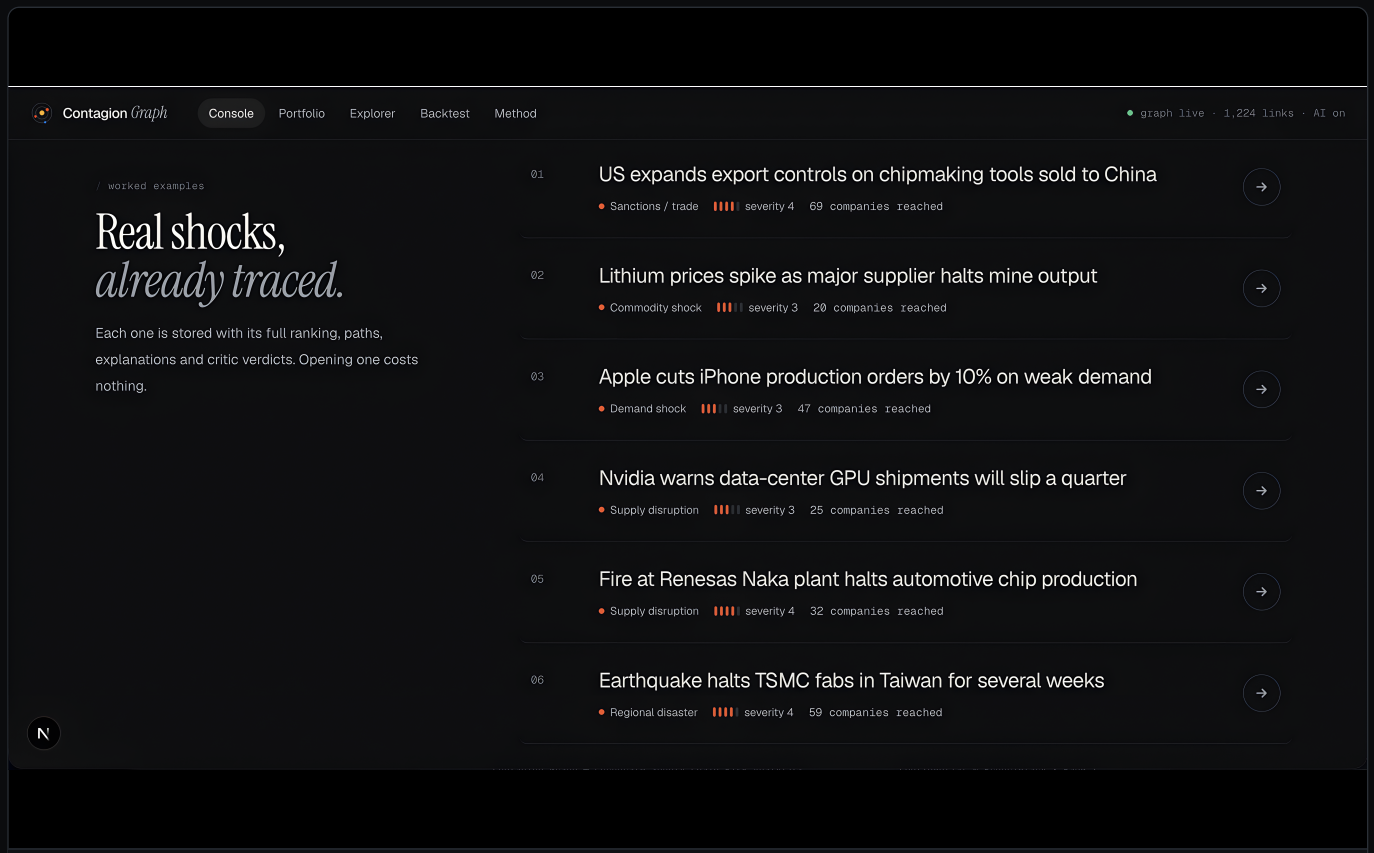

Figure 2: Execution Trace View — Run 0mfEJRZl on US chipmaking equipment export controls ($0.0019 AI cost). Displays completed pipeline chips, event taxonomy card (Severity 4/5, permanent sanctions), concentric radial network, and ranked exposure table with quoted evidence badges.

Technology Stack & *Empirical Validation.*

Layer	Primary Stack	Implementation Details
Frontend	Next.js 15, React 19, Tailwind v4	App Router with Turbopack, Instrument Serif editorial display, Geist Mono metrics.
Graph Engine	Cytoscape.js 3.30	Concentric radial layout, hardware-accelerated canvas, interactive hop ring filtering.
API Backend	FastAPI, Uvicorn, Python 3.12	Asynchronous REST endpoints hosted as Vercel Python Functions (`/api/py/*`).
Database	Neon Postgres (Pooled) & SQLite	Cloud relational persistence with connection pooling (`-pooler`); local SQLite parity.
Inference	Claude Haiku 5.5 (Batch & Live)	Batch API for offline filing extraction ($0.15 total spend); sub-cent structured live calls.
News Feeds	SEC EDGAR, GDELT & Google News	Direct regulatory crawler; global news polling with 15-minute response caching.

05 Empirical Backtest: 5 Historical Shocks

Model Architecture	5-Day Hit Rate vs SPY	Verifiable Evidence
Contagion Graph	70.0%	✓ 100% SEC Filings
AI Memory Baseline	**75.0%***	✕ None (Black Box)
Same-Sector Peers	66.0%	✕ None
Broad Market Universe	60.0%	✕ None

*AI Memory reflects historical training data contamination.

Production Economics

- **$0.0019 / Trace:** 3 lightweight Haiku calls (Parse, Explain, Critic).
- **$0.00 Replay Overhead:** SHA-256 headline caching.
- **Strict Safeguards:** $0.20/day cap & 8 runs/IP/hr rate limits.

06 Catalog of Pre-Indexed Shocks

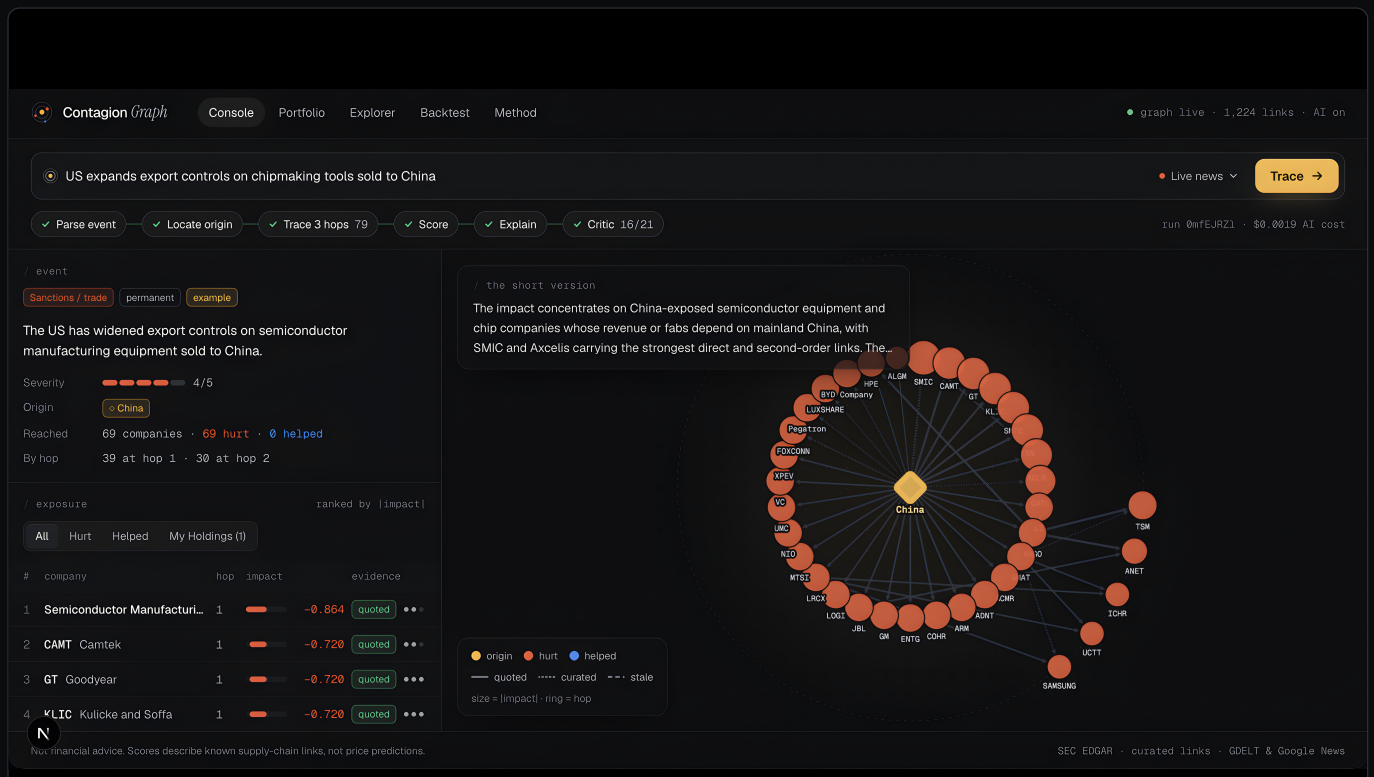

Figure 3: Catalog of Pre-Indexed Shocks — Stored worked examples (TSMC earthquake, Renesas plant fire, lithium mining halts, Apple order cuts). Enables instant public demonstration with zero incremental API spend.